Project Report

On

HOME IOT

Submitted To:

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BONAFIDE CERTIFICATE

This is to certify that the project report entitled **"Home IOT"** submitted by Sharad Pratap Singh, Naga Santosh of **University Institute of Computing**, is a Bonafide record of the work carried out under my/our guidance in partial fulfillment of the requirements for the award of the degree **Masters In Computer Application** in **General** of **Chandigarh University** during the academic year **2025-2026**.

The project work embodied in this report has not been submitted to any other University or Institution for the award of any degree or diploma.

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This project stands not merely as a piece of technical work, but as a testament to the collective effort, shared wisdom, and goodwill of all those who stood beside us in this.

Abstract

The **Centralized Routing Network System** is a network architecture model in which routing decisions are controlled and managed by a single central routing device or controller. Unlike traditional distributed routing systems where each router independently determines the best path for data transmission, centralized routing allows a central unit to maintain control over routing policies, traffic flow, and network management.

The primary objective of this project is to design, configure, and simulate a centralized routing network using networking concepts and tools such as Cisco Packet Tracer. The system connects multiple local area networks (LANs) through a central router that manages inter-network communication. Each network segment is assigned a unique IP address range, and the central router is configured to forward packets between these segments efficiently.

This project demonstrates how centralized control improves network management, simplifies configuration, enhances monitoring capabilities, and ensures better security policy enforcement. Through simulation and testing, the system verifies successful communication between different network segments using ping and connectivity tests.

The results show that centralized routing provides a structured and manageable approach to network design, especially for organizations where centralized administration and traffic control are required. The project highlights the practical implementation of routing concepts and provides a foundation for understanding modern centralized networking approaches such as Software Defined Networking (SDN).

Introduction

Computer networks and Internet of Things (IoT) technologies play a very important role in modern communication systems. IoT enables everyday devices to connect to the internet and communicate with each other, making homes smarter, safer, and more efficient. Smart home systems allow users to monitor and control appliances, sensors, and other devices through a network.

A Smart Home IoT Network connects various devices such as lights, fans, sensors, security systems, and other appliances through a central network infrastructure. These devices communicate using networking protocols and can be monitored or controlled remotely using computers or mobile devices.

In this project, a Smart Home IoT network is designed and simulated using Cisco Packet Tracer. The network includes IoT devices, a home gateway, wireless connections, and end-user devices. The system demonstrates how different IoT components communicate with each other through the network and how users can manage smart home devices efficiently.

The purpose of this project is to understand the working of IoT networks, device connectivity, and communication within a smart home environment using simulation tools. The project highlights how networking and IoT technologies can be integrated to create intelligent home automation systems.

Literature Review

Several studies and research works have explored the development of Internet of Things (IoT) technologies to improve automation, connectivity, and intelligent control in modern environments. IoT enables physical devices such as sensors, appliances, and controllers to communicate with each other through a network and exchange data over the internet.

Smart home automation is one of the most widely researched applications of IoT. Researchers have developed systems that allow users to remotely monitor and control household appliances such as lights, fans, security cameras, door locks, and temperature sensors using computers or smartphones. These systems improve convenience, enhance home security, and help in efficient energy management.

Many studies also highlight the importance of network simulation tools such as Cisco Packet Tracer for designing and testing IoT-based systems. Cisco Packet Tracer provides an interactive environment where IoT devices, home gateways, and network components can be configured and simulated. This helps students and researchers understand how smart devices communicate and operate within a network.

Research has shown that IoT-based smart home systems provide better automation, remote accessibility, and efficient device management. By integrating sensors, controllers, and networking technologies, smart homes can create an intelligent environment that responds to

user needs.

This project applies the concept of IoT-based smart home networking using Cisco Packet Tracer to simulate how different smart devices communicate within a home network and how users can monitor and control these devices effectively.

System Design

The system design of the Smart Home IoT Network includes several interconnected components that allow devices within a home environment to communicate and operate automatically. The network is designed and simulated using Cisco Packet Tracer to demonstrate how IoT devices interact within a smart home system.

1. Home Gateway / Wireless Router
   The home gateway acts as the central communication device of the smart home network. It connects all IoT devices and enables communication between them. The gateway also provides internet connectivity and allows users to control devices remotely.
2. IoT Devices
   IoT devices are the smart components used in the home environment. These may include smart lights, fans, doors, motion sensors, temperature sensors, and other automated appliances. These devices communicate with the home gateway through wired or wireless connections.
3. End User Devices
   End-user devices such as computers, tablets, or smartphones allow users to monitor and control smart home devices. Through these devices, users can manage the status of appliances and receive real-time updates from sensors.
4. Network Topology
   The topology of the smart home network consists of multiple IoT devices connected to a central home gateway. The gateway manages communication between devices and ensures that commands from the user are properly executed.

Design Architecture

The smart home IoT network architecture includes the following components:

• Home Gateway / Wireless Router
• IoT Devices (Smart lights, sensors, appliances)
• End User Devices (PCs or mobile devices)
• Wireless communication connections
• Internet connectivity for remote monitoring

In this architecture, IoT devices communicate with the home gateway, which acts as the central control unit of the smart home system. The gateway processes commands from the user and

sends instructions to the connected devices, allowing efficient automation and control within the home environment.

Implementation

The implementation of the Smart Home IoT Network is carried out using Cisco Packet Tracer. The simulation environment allows different IoT devices, networking components, and end-user devices to be connected and configured to demonstrate how a smart home system operates.

Steps for Implementation

1. Open Cisco Packet Tracer and create a new workspace for the smart home network design.
2. Add the required devices from the IoT and networking device list. These devices may include a Home Gateway (wireless router), IoT devices such as smart lights, fans, doors, motion sensors, temperature sensors, and user devices like a PC or smartphone.
3. Place the Home Gateway at the center of the network. This device will act as the main controller that connects and manages all IoT devices within the home network.
4. Connect the end-user device (such as a PC or laptop) to the Home Gateway using a wired or wireless connection so that the user can monitor and control the smart devices.
5. Configure the IoT devices and register them with the Home Gateway. Each device is assigned a unique identifier so that it can communicate with the gateway.
6. Configure wireless connectivity so that IoT devices can connect to the home network through the gateway.
7. Set up the IoT monitoring interface in Cisco Packet Tracer to control and observe the status of connected devices.
8. Test the system by sending commands from the user device to IoT appliances such as turning lights on/off or activating sensors.

The implementation demonstrates how different IoT devices communicate through a home gateway and how users can control smart home appliances through a centralized network system within Cisco Packet Tracer.

Working

The Smart Home IoT Network operates by connecting various smart devices to a central home gateway that manages communication between devices and users. The gateway acts as the main controller that allows devices to communicate within the network.

1. When a user sends a command from a computer or mobile device, the request is transmitted through the home network to the Home Gateway.
2. The Home Gateway processes the command and identifies the specific IoT device that needs to perform the action.
3. The command is then forwarded to the corresponding IoT device, such as a smart light, fan, or sensor.
4. The IoT device executes the command and sends its status back to the gateway.

5. The user can monitor the device status and control multiple smart devices through network interface.

In this system, all smart devices communicate through the Home Gateway, which manages the interaction between devices and users. This allows efficient automation, monitoring, and control of household appliances within the smart home environment.

Testing and Results

After completing the configuration of the Smart Home IoT Network in Cisco Packet Tracer, testing was performed to ensure that all IoT devices were properly connected and functioning within the network.

Testing Methods

• Device Control Test
• Sensor Monitoring Test
• Network Connectivity Test

Results

• All IoT devices were successfully connected to the Home Gateway.
• The user device was able to monitor and control smart devices such as lights, fans, and sensors through the network interface.

• Commands sent from the user device were correctly executed by the IoT appliances.
• Sensor devices were able to send status information to the Home Gateway in real time.

These results confirm that the Smart Home IoT Network functions correctly and allows efficient communication and control of devices within the simulated home environment

Applications

Smart Home IoT systems have a wide range of applications in modern living environments. These systems use internet-connected devices to automate and control various household activities.

• Home Automation
Smart devices such as lights, fans, air conditioners, and other appliances can be automatically controlled through a network.

• Home Security Systems
IoT-based smart homes can include security cameras, motion sensors, and smart door locks to improve home safety.

• Energy Management
Smart home systems help monitor and control energy usage by automatically managing appliances and lighting.

• Remote Monitoring and Control
Users can monitor and control their home devices from anywhere using computers or mobile devices connected to the internet.

• Smart Living Environments
IoT technology helps create comfortable and intelligent living environments by integrating various smart devices within a home network.

These applications demonstrate how IoT technology can improve convenience, security, and efficiency in modern homes.

Advantages

The Smart Home IoT Network offers several advantages in modern living environments:

• Convenience and Automation
Smart devices allow users to automate daily household activities such as controlling lights, fans, and appliances.

• Remote Access and Control
Users can monitor and control their home devices from anywhere using internet-connected devices such as smartphones or computers.

• Improved Home Security
IoT systems can integrate security devices such as smart cameras, motion sensors, and smart locks to enhance home safety.

• Energy Efficiency
Smart devices help reduce energy consumption by automatically managing lighting and appliances.

• Easy Monitoring
Users can easily monitor the status of different home devices and sensors through a centralized interface.

Limitations

Despite its benefits, Smart Home IoT systems also have some limitations:

• Security and Privacy Risks
If not properly secured, IoT devices may be vulnerable to cyber attacks or unauthorized access.

• Dependence on Internet Connectivity
Many smart home functions require internet access, and system performance may be affected during network failures.

• Initial Setup Cost
Installing smart home devices and network components may require an initial investment.

• Device Compatibility Issues
Different IoT devices from various manufacturers may sometimes face compatibility or integration issues.

Future Scope

The future scope of Smart Home IoT systems is very promising as technology continues to evolve. With advancements in networking and intelligent systems, smart homes are expected to become more efficient, secure, and automated.

• Integration with Artificial Intelligence (AI)
Future smart home systems can use artificial intelligence to automatically learn user behavior and control devices intelligently.

• Advanced Home Automation
More household appliances and electronic devices will be integrated into IoT networks to create fully automated living environments.

• Improved Security Systems
Future smart homes will include advanced security features such as facial recognition, smart surveillance cameras, and automated alert systems.

• Energy-Efficient Smart Homes
IoT systems will help optimize energy consumption by automatically managing lighting, heating, and other appliances.

• Integration with Smart Cities
Smart home systems may become part of larger smart city infrastructures, allowing better connectivity between homes, transportation systems, and public services.

These advancements will make smart home IoT systems more intelligent, reliable, and widely used in modern society.

Conclusion

The Smart Home IoT Network project demonstrates how Internet of Things technology can be used to automate and control devices within a home environment. The project successfully designed and simulated a smart home network using Cisco Packet Tracer.

Through this project, various IoT devices were connected to a home gateway, allowing

communication between smart appliances and user devices. The simulation showed how users can monitor and control home devices such as lights, sensors, and other appliances through a network interface.

The results highlight how IoT technology can improve convenience, security, and efficiency in modern homes. The project also helped in understanding important concepts such as IoT device connectivity, network communication, and smart home automation.

Overall, the Smart Home IoT Network provides a practical demonstration of how networking and IoT technologies can be integrated to create intelligent and automated living environments.

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